

Data Communication & Computer Networks (CS24305) — Full Course Master Book

Complete End-to-End B.Tech CSE 5th Semester Study Book & PYQ Bank

EXECUTIVE MASTER STUDY GUIDE & PROTOCOL BANK

Data Communication & Computer Networks (CS24305)

Birla Institute of Technology, Mesra | B.Tech CSE 5th Semester (NEP 2024–25 Scheme)

Complete Course Structure & Protocol Matrix

Module	Layer & Focus	Key Formulations, RFCs & Protocols
Module I	Physical Layer & Signals	Nyquist & Shannon Capacity, Transmission Impairments, Line Coding (Manchester, AMI), Multiplexing (FDM, TDM, WDM)
Module II	Data Link Layer & MAC	Framing, CRC-32 Polynomial Derivations, Sliding Window ARQs (Go-Back-N, Selective Repeat), CSMA/CD & CSMA/CA
Module III	Network Layer & Routing	IPv4/IPv6 Header Analysis, CIDR Subnetting Calculations, Dijkstra Shortest Path, Bellman-Ford, OSPF, BGP-4
Module IV	Transport Layer & Protocols	TCP 3-Way Handshake, TCP State Machine, AIMD Congestion Control, Slow Start, Fast Retransmit, UDP Sockets
Module V	Application Layer & Security	DNS Hierarchical Resolution, HTTP/1.1 vs HTTP/2/3, TLS 1.3 Handshake, RSA Asymmetric Encryption, AES

Exam Preparation & High-Yield Strategy

This publication-grade master book consolidates all 5 modules with formal mathematical channel capacity proofs, step-by-step worked subnetting & CRC numericals, packet header diagrams, and model answers to BIT Mesra end-semester examination questions.

Module I: Data Communications & Networking Overview — Topics Covered

1. Five Components of Data Communication
2. Network Topologies (Mesh, Star, Bus, Ring)
3. OSI 7-Layer vs. TCP/IP 4-Layer Architecture
4. Transmission Impairments (Attenuation, Delay, Noise)
5. Nyquist Maximum Data Rate Theorem
6. Shannon Channel Capacity Theorem & SNR (dB)

1. Layered Network Models: OSI vs. TCP/IP

OSI Layer	Protocol Data Unit (PDU)	Key Responsibilities	TCP/IP Equivalent
7. Application	Messages / Data	Network virtual terminal, HTTP, FTP, SMTP, DNS	Application Layer
6. Presentation	Formatted Data	Translation, Encryption (SSL/TLS), Compression	
5. Session	Dialog Tokens	Session establishment, Dialog control, Sync checkpoints	
4. Transport	Segments (TCP) / Datagrams (UDP)	Port addressing, Segmentation, Flow/Error/Congestion control	Transport Layer
3. Network	Packets	Logical IP addressing, Subnet routing, Path determination	Internet Layer
2. Data Link	Frames	Physical MAC addressing, Framing, Error detection (CRC)	Network Access Layer
1. Physical	Bits	Bit transmission over copper/fiber/radio, signal levels	

2. Channel Capacity Theorems

2.1 Nyquist Theorem (Noiseless Channel)

$$C = 2B \log_2(M) \text{ bps}$$

where B is channel bandwidth in Hertz and M is the number of discrete signal voltage levels.

2.2 Shannon Theorem (Noisy Channel)

$$C = B \log_2(1 + \text{SNR}) \text{ bps}$$

where $\text{SNR} = \frac{P_{\text{signal}}}{P_{\text{noise}}}$. Given $\text{SNR}_{\text{dB}} = 10 \log_{10}(\text{SNR}) \implies \text{SNR} = 10^{(\text{SNR}_{\text{dB}}/10)}$.

BIT Mesra Mid-Sem Exam Numerical (8 Marks)

Problem: A telephone line has a bandwidth of 4 kHz and a signal-to-noise ratio of 30 dB. Calculate the theoretical Shannon maximum channel capacity.

Solution:

$$\text{SNR}_{\text{dB}} = 30 \implies 10 \log_{10}(\text{SNR}) = 30 \implies \text{SNR} = 10^3 = 1000$$

$$C = B \log_2(1 + \text{SNR}) = 4000 \times \log_2(1001) \approx 4000 \times 9.967 = \mathbf{39,869 \text{ bps} \approx 39.87 \text{ kbps}}$$

Module II: Transmission Media & Signal Encoding — Topics Covered

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|---|--|
| 1. Guided Media (Twisted Pair, Coax, Optical Fiber) | 2. Digital Signal Encoding: NRZ, Manchester, AMI |
| 3. Analog Modulation: ASK, FSK, PSK, QAM | 4. Pulse Code Modulation (PCM) & Quantization |

1. Digital Signal Encoding Techniques

Encoding Scheme	Rule / Signal Transition	Key Advantages & Drawbacks
NRZ-L (Level)	Bit 0 = High level, Bit 1 = Low level	Simple; susceptible to DC baseline wander and clock drift on long runs of 0s/1s.
NRZI (Invert)	Bit 1 = Transition at start; Bit 0 = No transition	Solves sync for consecutive 1s, but fails on strings of 0s.
Bipolar-AMI	Bit 0 = Zero voltage; Bit 1 = Alternating +V and -V	Zero DC component; easy error detection; fails sync on consecutive 0s.
Manchester (Ethernet)	Bit 0 = High-to-Low transition; Bit 1 = Low-to-High transition	Self-clocking (transition at middle of every bit); requires $2\times$ bandwidth.
Diff. Manchester	Transition in middle; Bit 0 = Transition at start; Bit 1 = No transition at start	Differential noise immunity; self-clocking; standard in Token Ring (IEEE 802.5).

2. Analog Modulation & 16-QAM Constellation

Quadrature Amplitude Modulation (QAM) modulates both amplitude and phase simultaneously. In 16-QAM, each transmitted constellation point represents **4 bits** ($\log_2(16) = 4$), enabling high spectral efficiency.

Module III: Error Control & Data Link Protocols — Topics Covered

1. Cyclic Redundancy Check (CRC-16/32 Modulo-2)
2. Hamming Distance & (7, 4) Error Correction
3. Sliding Window ARQ: Stop-and-Wait, GBN, SR
4. HDLC Protocol & Bit Stuffing Mechanics
5. Frequency and Time Division Multiplexing (FDM/TDM)

1. Cyclic Redundancy Check (CRC)

Given data word $D(x)$ of k bits and generator polynomial $G(x)$ of degree n :

- Append n zero bits to $D(x)$ to form dividend $D(x) \cdot 2^n$.
- Perform binary Modulo-2 division (XOR) by generator polynomial $G(x)$.
- The n -bit remainder $R(x)$ is the Frame Check Sequence (FCS). Transmit $T(x) = D(x) \cdot 2^n \oplus R(x)$.

2. Sliding Window ARQ Protocols Comparison

Protocol	Sender Window (W_s)	Receiver Window (W_r)	Efficiency (η)	Retransmission Behavior
Stop-and-Wait	1	1	$\frac{1}{1+2a}$ where $a = \frac{T_{prop}}{T_{trans}}$	Retransmits single frame upon timer expiry.
Go-Back-N (GBN)	$2^k - 1$	1	$\frac{W_s}{1+2a}$	Discards out-of-order frames; retransmits entire window from lost frame.
Selective Repeat	2^{k-1}	2^{k-1}	$\frac{W_s}{1+2a}$	Buffers out-of-order frames; retransmits only the damaged frame (NAK).

Module IV: Switching, Cellular Networks & LANs — Topics Covered

1. Circuit Switching vs. Packet Switching
2. Cellular Frequency Reuse ($N = i^2 + ij + j^2$)
3. IEEE 802.3 Ethernet & CSMA/CD Backoff
4. IEEE 802.11 Wi-Fi & CSMA/CA (RTS/CTS)
5. Virtual LANs (VLANs & IEEE 802.1Q)

1. Ethernet CSMA/CD vs. Wi-Fi CSMA/CA

Feature	IEEE 802.3 Ethernet (CSMA/CD)	IEEE 802.11 Wi-Fi (CSMA/CA)
Mechanism	Carrier Sense Multiple Access with Collision Detection	Carrier Sense Multiple Access with Collision Avoidance
Wireless Challenge	Transceiver can detect collision while sending.	Cannot detect collision while transmitting (RF blindness) $\implies$ Must avoid collisions.
Collision Recovery	Emits 32-bit Jam signal; Binary Exponential Backoff ($[0, 2^k - 1] \times 51.2\mu s$).	Interframe Spaces (DIFS, SIFS); RTS/CTS handshaking resolves hidden terminals.

Module V: Internetworking, TCP/IP & Routing — Topics Covered

1. IPv4 Addressing, Classless CIDR & VLSM Subnetting
2. TCP 3-Way Handshake & Congestion Control
3. Dijkstra Shortest Path (OSPF) & Bellman-Ford (RIP)
4. Application Protocols: DNS, DHCP, HTTP/2, SMTP

1. TCP Congestion Control Mechanics

TCP maintains a congestion window ($cwnd$) and slow-start threshold ($ssthresh$):

- **Slow Start:** $cwnd$ starts at 1 MSS and doubles every RTT ($cwnd = cwnd \times 2$) exponentially until $cwnd \geq ssthresh$.
- **Congestion Avoidance:** $cwnd$ increases linearly by 1 MSS per RTT (Additive Increase).
- **Triple Duplicate ACKs (Fast Retransmit & Recovery):** $ssthresh = cwnd/2$, $cwnd = ssthresh + 3$ MSS, retransmit lost segment immediately.
- **Timeout:** $ssthresh = cwnd/2$, $cwnd = 1$ MSS, re-enter Slow Start.

BIT Mesra VLSM Subnetting Problem (10 Marks)

Problem: Given network block `192.168.1.0/24`, design subnets for Dept A (100 hosts), Dept B (50 hosts), and Dept C (25 hosts).

Solution:

- **Dept A (100 hosts):** Needs $2^7 - 2 = 126$ IPs $\implies$ /25. Subnet: `192.168.1.0/25` (Range: `.1` to `.126`, Broadcast: `.127`).
- **Dept B (50 hosts):** Needs $2^6 - 2 = 62$ IPs $\implies$ /26. Subnet: `192.168.1.128/26` (Range: `.129` to `.190`, Broadcast: `.191`).
- **Dept C (25 hosts):** Needs $2^5 - 2 = 30$ IPs $\implies$ /27. Subnet: `192.168.1.192/27` (Range: `.193` to `.222`, Broadcast: `.223`).